

Lab 5: Car Design Project

Olivia Santiago (obs23), Anish Dhawan (akd82) & Charles Wall-Davis (caw362)

ENGRI 1170, Lab Section 441

Dominika Swedek

Nov. 30, 2023

Introduction

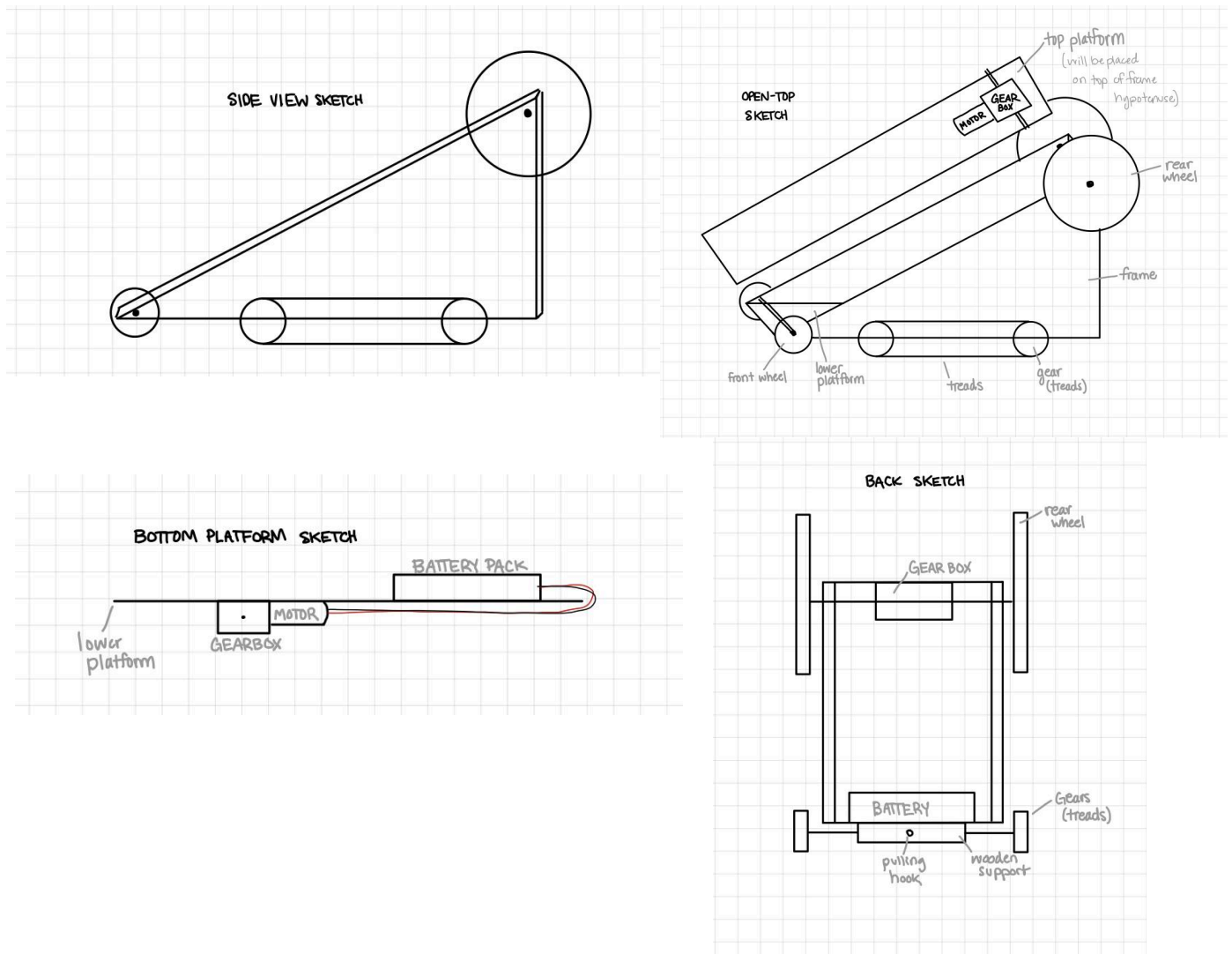
The specific purpose of Lab 5: Car Design Project is to use a thoughtful design process and demonstrate good engineering. Not to mention, in this lab, we were tasked to design and manufacture a small car powered by a DC motor and battery pack, which will be tested in a competition with three events. Our objective was to build a car that'll deliver high pulling force, or the maximum pulling force per weight, as well as maximum speed, variables dependent on the car body, tires, gear system, and so on. For data collection, the methods utilized were to record the car's weight from a scale and, having an already known distance, record the time the car takes to go from point A to point B. From our analysis/calculations, the information that'll be obtained are the coefficient of friction, the predicted maximum velocity, maximum pull force, and the competition calculated values of maximum velocity along with the maximum pull force.

Development of Design

To start, our car was originally designed to have a structure like that of a tank. After looking at the past competition cars, we wanted to implement treads (tracks) into our vehicle because it'll look nice for aesthetics, and that idea will give us better results for the tug-of-war along with the pulling force events for the competition. Wanting to have a structure similar to a tank, that'll mean that our car would be a square shape. But, in the end, we chose to have a triangular shape because it's a great blend between strength plus speed, it's great aerodynamics wise, and it's popular among the past students. For speed, having a triangular shape, we decided that our car should have a big wheel above the treads (flippable). Yet for the big wheel to move the car, front wheels, which would be moved from both the treads and big wheels, were added.

In the Design Presentation Slides, from November 2 to November 8, we were to refine measurements, order materials, and start manufacturing the car. From November 9 to the 15th,

we would finish manufacturing the car and see its performance. For November 16 to the 17th, we would finish testing and final modifications based on previous trials. Then, on November 18, it would be competition day. Hence, we would complete last minute additions and attend prior to start for troubleshooting. This is the project plan/schedule in which we tried to follow, but we had trouble doing so. This discrepancy's cause was from the car's structure. Originally, we wanted to glue the wood pieces together, but the structure would still break after a lot of pressure is applied on it. From that, we decided to connect the pieces together using mortise and tenon joints (notch), which was very slow due to it requiring accurate as well as precise measurements.



Manufacturing Process

In our manufacturing process, what went well is that we were able to finish our essential tasks relatively quickly, aside from connecting the overall structure. Not to mention, when we tested the car for the first time, it was able to move easily. Hence, we had no fatal flaws in our design, the flaws requiring us to change our structure as a whole. For our design, we mentioned this before, we decided to use notches to connect our wood, so it'll be more secure. Additionally, deviating from above, our smaller (front) wheel was positioned in a way such that it'll work with the bigger wheel, not the treads. This is due to the final structure not allowing the smaller wheel to work on both methods of transport and the fact that the treads can move without any support. For our manufacturing plan, it didn't change at all because we utilized our lab hours and office hours, as we said we'd do in our Team Schedule slide in the Design Presentation Slides.

Table 1: Bill of Materials (BOM)

Materials	Costs
Wood (body)	Free
Motors (2)	Free
Pololu Wheel 90x10mm Pair	\$7.95
Pololu Wheel 40x7mm Pair	\$4.95
No 100 Track and Wheel Set	\$10.50
Gear Boxes	Free
Rocks!	Free

Total Price of Materials < \$23.40

Analysis of Performance

Sample Calculations for Predictions

Measured Coefficients of Friction

$$\mu_s = \tan\theta$$

$$\mu_s (\text{treads}) = \tan(40^\circ) = 0.8390996312$$

$$\mu_s (\text{wheels}) = \tan(5^\circ) = 0.0874886635$$

In physics, the equation for the force of friction is $F_s = \mu_s N$, where N is the normal force from the ramp. Having the cart placed on the ramp, $F_s = mg\sin\theta$ and N is equal to $mg\cos\theta$. Hence, $mg\sin\theta = \mu_s mg\cos\theta$. So, solving for μ_s , the equation is $\mu_s = \tan\theta$. Or, it's the equation that is shown above.

Maximum Velocity (Top Speed)

$$F_{\text{friction}} = ((.6)(.55\text{kg})(9.81 \text{ m/s}^2)(.25\text{m}/2))/(.25\text{m}) = 1.61865 \text{ N}$$

$$M_{\text{motor}} = ((1.61865 \text{ N})(.045\text{m}))/(.127) = 5.74 \times 10^{-3} \text{ Nm}$$

$$\omega = (((5.74 \times 10^{-3} \text{ Nm}) - (.01362 \text{ Nm}))(14000 \text{ RPM})) / (-.01362 \text{ Nm}) = 13999 \text{ RPM} = 1466.032474 \text{ rad/s}$$

$$V_{\text{wheel}} = ((1397.694572 \text{ rad/s})(.045 \text{ m})) / (.127) = 5.194603255 \text{ m/s}$$

For the determination of V_{wheel} , we first assumed that the coefficient of friction is .6. From above, to find V_{wheel} , we must first find F_{friction} , M_{motor} , and ω . To find F_{friction} , we used the moment method to find the normal force. However, since it's F_{friction} we're finding, we must multiply that by the coefficient of friction since it's $F_{\text{friction}} = \mu_{\text{friction}} N$. To find M_{motor} , we multiplied the F_{friction} by the wheel's radius over the gear ratio and M_{motor} and the GR have an inverse relationship. Then, we used this value to find ω by subtracting it and dividing it over the maximum specifications shown

in the Design Project Description document. For V_{wheel} , it was simple multiplication as we converted w to rad/s, or s^{-1} as radian (rad.) is unitless, and multiplied it by the wheel's radius.

Maximum Pulling Force (Pull Force)

$$F_{\text{pulling}} = (T_{\text{stall}} \times GR) / (r_{\text{wheel}})$$

$$F_{\text{pulling}} = ((1.362 \times 10^{-2} \text{ Nm})(344.2)) / (.018 \text{ m}) = 260.44466667 \text{ N}$$

To find F_{pulling} , we used the equation shown above as torque is the rotational analogue of linear force, so it's $rF\sin\theta$. Having the radius be in the torque equation, it's necessary to divide the stall torque by the radius of the wheel, to get $F\sin\theta$. With this, for the final step to get F_{pulling} , we have to multiply $F\sin\theta$ by the GR ratio because the pulling force is entirely dependent on that, it has a positive (~~inverse relationship~~) correlation. Hence, as the gear ratio increases, F_{pulling} will increase.

Average Velocity (Competition)

$$\text{velocity} = (\text{displacement}) / (\text{change in time})$$

$$\text{velocity} = (12\text{m}) / (126\text{s}) = 0.09523809523 \text{ m/s}$$

From above, the equation for velocity is displacement over the change in time. With the car being at 0 m when the timer was about to start at (0 sec), velocity was calculated from the racetrack's length over the amount of seconds elapsed to travel it. So, 12m over the seconds.

Maximum (Best) Pull Force

$$1 \text{ lbf} = 4.44822 \text{ N}$$

$$0.95 \text{ lbf} = 4.225811 \text{ N}$$

From above, to find Newtons, from lbf, you must multiply the lbf by 4.44822 N. So, to convert our Maximum (Best) Pull Force to N, we must multiply it by 4.44822 N. The origin for this equation is that $1 \text{ kg} = 2.2 \text{ lb}$. So, in turn, $2.2 \text{ lbf} = 9.8 \text{ N}$. Dividing by 2.2, $1 \text{ lbf} = 4.44822 \text{ N}$.

Against the performance, our predicted results were highly inaccurate. Part of this is because we used the tracks (treads) instead of the larger wheels in the Speed Test. Also, these tracks, from before, were connected to Gearbox D. So, in turn, that would result in a much slower vehicle, being that it took two minutes for the car to pass the finish line. For the Pull Force, our theoretical calculation was around 60 times that of the actual. This could be from the car moving slow, from the tracks, causing less force to be exerted forward (recorded). Though, this assumption is due Gearbox D, meaning that the speed the car displayed is wrong. So, it must've been a manufacturing issue. This is an error that'll also be stated down below too.

For the errors, it could be that the Gearboxes weren't built correctly (causing a poor performance), something got in between the gears of the Gearboxes (tape and/or hot glue), and the axle's hot glue interfered with the wooden structure of the car. In our calculations, the assumptions made were that: Gearbox D was utilized for the Pull Force, Gearbox A was used in the Speed Test, 550 g (.55 kg) was the car's weight, the coefficient of friction was .6, the stall torque as well as the no load were the maximum outputs of the car's torque and RPM. Not to mention, the gravity is 9.81 m/s^2 , the radius of the tracks were .018 m, the big wheels .25 m,

Gearbox A is 12.7, and Gearbox D is 344.2. In actuality, the car’s weight is 724.7 g (.7247 kg).

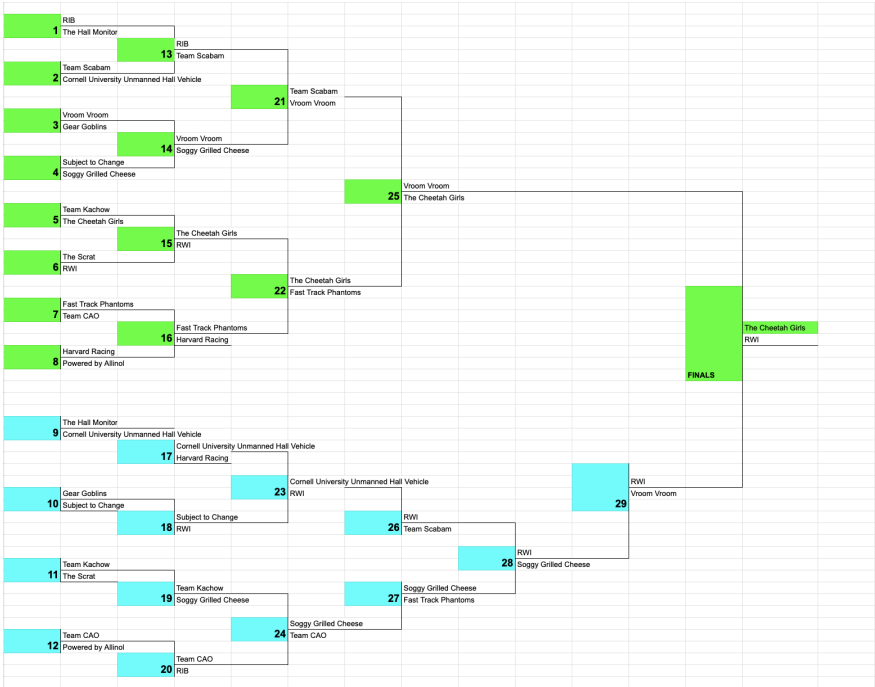
Table 2: Final Results From Competition Day (Pull Test)

Team Name	Trial 1				Trial 2			Bests		
	Weight	Av Pulling Force	Peak	Normalized	Av Pulling Force	Peak	Normalized	Best Pull Force	Best Peak	Best Normalized (lbf/g)
CaO	724.7	0.95	1.76	0.00131089	0.62	1.25	0.0008555264247	0.95	1.76	0.00131089

Table 3: Final Results From Competition Day (Speed Test)

Team Name	Trial 1	Trial 2	Best
Team CAO	DNF	2:06	126 s

Figure 1: Final Results From Competition Day (Tug of War)



During the competition, what went well is our car's performance in the Pull Test and Tug of War events. In the Pull Test, we originally expected our car to place considerably high because our car was the heaviest when compared to all the other competition cars and it used treads connected to Gearbox D (344.2), giving it a relatively high pulling force (from calculations). With this, our goal was met as our car placed 5th in the best overall peak. For the Tug of War, we thought our car would win the entire event since our car weighed 724.7 g, way more than any other car. Yet, even though it lost relatively early, it still had the same number of wins compared to any other tread, or track, centered (focused/oriented) car (2 wins). For example, Team Scabam.

During the competition, what didn't go as expected was the Speed Test. During the Speed Test, our Trial 1 was a DNF because our car hit the wall. This is because our car's front wheels were crooked due to the misalignment of the gearbox + motor, or it was from the wheels being titled when it was hardening onto the axle by the hot glue gun. Resulting in the car not going straight as it moved and hitting the right end of the wall. For Trial 2, we were able to get a recording of time. Yet, during that time, the car moved using its tracks connected to Gearbox D. Making the time be 126 seconds (An Increasing Gear Ratio (GR) = A Slower Velocity (Speed)).

Given its final performance, the improvements to our car are as follows: use plastic material, or laser cut material from CAD, use a different method of fastening the wheels onto the axle instead of hot glue, use Gearbox A or B instead of D for the treads, and so on. During our design process, as said above, most of our time was spent building the car's structure. Also, being wood, it took even longer to cut than plastic. From this, it would be better to use plastic instead of wood since it cuts down the mass of the vehicle by a significant margin and it's more adjustable. As wood is more thick, it'll take much more time to mold it into a targeted structure.

Or, we could use material that was already laser cut from CAD since that saves a lot of time, minus understanding the CAD program in general. Also, as CAD would make holes to match our vehicle's targeted direction as it moves, it would fix the problem of it crashing into the wall for our Speed Test. Stated before, using hot glue might not be a reliable way of fastening the wheels onto the axle as the wheels could be crooked from the amount of hot glue applied. So, another method is in order. Finally, for the Gearboxes, it would be better to use A or B for the treads as our 1st loss in the Tug of War was from the opposing car's acceleration being better.